TECHNOLOGY ETHICS • COMPLETE STUDY NOTES

Set 8: Recurring CAT Topics (Professional Practice & Emerging Tech)

Occupation vs Profession, Distinguishing Features of Professional Bodies, Whistleblowing, Conflict of Interest, AI in Education & Environmental Ethics

PART 8

Recurring in CATs but not in your slides

2 sections, about 1 h 35 min of reading.

Profession versus occupation, whistleblowing and conflict of interest

~1 h 00 min to read | High priority | 18 questions

Source: not in your slides. Built from the question bank, the 2011-2018 papers, the CAT papers and the three solution files. The only slide material used is the model-rules wording in Set 4, slides 11-25 (whistle blowing and conflict of interest clauses).

Not in your slides. The lecturer never taught profession versus occupation, whistleblowing or conflict of interest as separate slide topics, but they come up in almost every CAT and exam: profession and occupation in 2016, 2017, CAT 2021-22 and the 2021-22 exam; whistleblowing in 2011/12 and 2013/14; conflict of interest in 2011/12, 2013/14 and two CAT 2 papers. Treat this as a full topic. Where a passage is general knowledge and not from a slide or a solution file it carries the label **beyond the slides**. Where the wording is a Set 4 model rule, the text says so.

How the Must/Should tags work here. The rule is: Must = phrase is in the slides and in a past paper or solution; Should = slides only. Because most of this section is not in the slides, almost every marker below is tagged by what the pool shows. Where a phrase is not in the slides but appears in all the solution files (for example "specialized knowledge", "code of ethics"), it is tagged Must and the row says why. Phrases that appear only in one paper or one solution are tagged Should.

Profession versus occupation

An **occupation** is any activity a person does to earn a living. It covers every kind of paid work: driving a taxi, selling goods, cleaning, farming, clerical work, and also medicine or engineering. A **profession** is a special kind of occupation. It is a vocation that needs specialized knowledge, long training and formal qualifications, and it is organised and regulated by a professional body that sets standards of competence and a code of ethics. All professions are occupations. Not all occupations are professions.

The solution files give the same core split in three places. The CAT 2021-22 solution says a profession is "a paid occupation that requires specialized knowledge, training, and often formal qualifications, such as medicine or law" while an occupation is "a broader term that encompasses any activity or job that a person does to earn a living". The 2020-21 exam solution adds that a profession is "often regulated by a professional body or association" and that professions involve "a higher level of expertise, ethical standards, and professional responsibilities". The 2021-22 exam solution compares on four points: level of specialization, formal education, autonomy, and adherence to a code of ethics.

The points of difference

Point	Profession	Occupation
Knowledge	Specialized, theoretical and technical body of knowledge	Skills that can be learned on the job; no deep theory required
Education and training	Long formal education (a degree), then supervised practice	Short training, apprenticeship or none
Entry and licensing	Qualification, registration or licensing before you may practise (for engineers, registration with the ERB and UIPE)	Anyone may enter; no licence needed
Regulation	Governed by a professional body that sets standards and disciplines members	Governed by the employer and general law only
Code of ethics	Binding code, breach can lead to sanctions or deregistration	Usually no profession-wide code, only company rules
Autonomy	Practitioner uses independent professional judgment	Follows instructions and procedures set by the employer
Responsibility	Duty to the public and personal accountability for results	Accountable mainly to the employer
Purpose	Service to society as well as earning a living	Mainly earning a living

Examples

The 2021-22 exam solution uses two pairs. **Nursing versus housekeeping:** nursing needs specialized education, licensure, a professional code of ethics and autonomy in decisions about patient care. Housekeeping is cleaning work done without the same formal education, specialized training or code. **Law versus retail sales:** law has extensive formal education, specialized training, licensure, a code of conduct and autonomy in giving legal advice. Retail sales is selling products without those requirements. Other pairs that work: engineer versus bricklayer, doctor versus shop attendant, accountant versus cashier.

Watch out. Do not say that an occupation has no skill or no value. The difference is in the features above, not in how important the job is. A skilled welder has an occupation. A welding engineer registered with the ERB who signs off a design is practising a profession.

Q: `ethics-cat-1 Q1a · 3 marks` **Explain the difference between an occupation and a profession.**

Q: `CAT 2021-22 Q1 (iii) / 2016 Q1c · part of 12 marks` **Differentiate between the following: ... Profession and Occupation.**

Q: `Exam 2021-22 Q1b · marks not printed` **With aid of examples briefly describe how profession differs from a mere occupation.**

What qualifies engineering as a profession

Engineering qualifies as a profession because it has every feature in the table above. The question asks you to show that the features apply to engineering, so state the feature and attach engineering to it.

1. **Specialized body of knowledge.** Engineering rests on mathematics, science and design methods that take years to master.
2. **Long formal education.** A university degree is required, followed by supervised experience.
3. **Registration and licensing.** In Uganda, graduate engineers register with the Engineers Registration Board (ERB) and join the Uganda Institute of Professional Engineers (UIPE). Graduation alone does not make you registered. The ERB is the statutory regulator that registers engineers; the UIPE is the professional membership body. They have different functions (the question bank statement that they "serve the same function" is false), and registration or membership is by application, not automatic. The question bank has a statement that qualified engineers are "automatically registered on graduation": it is false.
4. **A professional body.** UIPE and international bodies (IEEE, ASCE, ASME) set standards and represent members.
5. **A code of ethics.** Engineers are bound by canons that put the safety, health and welfare of the public first.
6. **Autonomy and judgment.** Engineers make independent technical decisions, such as approving a design, that others rely on.
7. **Public service and public trust.** Engineering work affects lives (bridges, power, communications), so society trusts and expects accountability from engineers.
8. **Accountability and discipline.** Breach of the code can lead to disciplinary action, loss of registration, and legal liability.

Q: `2017 Q5F · 3 marks` **What qualifies Engineering as a profession?**

MARKER WORDS: PROFESSION VERSUS OCCUPATION

Say it like this:

> A profession is an occupation that requires specialized knowledge, training and formal qualifications, is regulated by a professional body and is bound by a code of ethics; an occupation is any work a person does to earn a living.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
specialized knowledge	defining a profession (CAT 2021-22 solution)	MUST
formal education / formal qualifications	saying what separates the two (exam 2021-22 solution)	MUST
regulated by a professional body	explaining who sets standards	MUST
code of ethics	listing what a profession has and an occupation lacks (slides use this phrase heavily)	MUST
autonomy	describing independent judgment (exam 2021-22 solution)	SHOULD
licensure / registration	describing entry to practice	SHOULD
earn a living	defining an occupation	MUST
public trust / public welfare	showing engineering serves society	SHOULD

▶ Recall drill: write the definition from memory, then click each blank to check

FLASHCARDS: CLICK TO FLIP

Define an occupation.
Any activity or job that a person does to earn a living, including both professional and non-professional work.

Define a profession.
A paid occupation that requires specialized knowledge, training and formal qualifications, regulated by a professional body, with ethical standards and professional responsibilities.

The four points on which the 2021-22 solution compares a profession with an occupation.
Level of specialization; formal education; autonomy; adherence to a code of ethics.

Two example pairs of profession versus occupation.
Nursing versus housekeeping; law versus retail sales.

Characteristics that distinguish a professional body

A **professional body** is an organisation of qualified practitioners of one profession. The Uganda Institute of Professional Engineers (UIPE) is the engineering example in the papers. Its purpose is to regulate and serve its profession. Other social organisations, such as clubs, churches or charities, may have members and rules, but they do not control entry to a trade or license people to practise. The solution files list five distinguishing features.

1. **Regulation of standards.** The body establishes and maintains standards of competence, conduct and ethics for the profession.
2. **Certification, licensing and registration.** The body, or a board it works with (the ERB for engineers), oversees who is qualified to practise. People who do not meet the qualifications are excluded.
3. **Continuing professional development (CPD).** The body offers training, seminars and courses so members keep learning and stay competent.
4. **Advocacy and representation.** The body speaks for its members and for the profession to government, regulators and other stakeholders, and it advises on policy and law.
5. **Ethics and discipline.** The body publishes a code of ethics and runs disciplinary procedures, so misconduct can be punished by warning, suspension or removal from the register.

beyond the slides Other features you can add if the question asks for more: it accredits university programmes; it sets entry exams; it is self-governing; it protects the public by guaranteeing a minimum standard; it gives members a professional title and identity.

Q: QB B1 · 4 marks **Give four characteristics that distinguish a professional body from other social organizations.**

Q: Exam 2020-21 Q1C · marks not printed **Which distinguishing features do professional bodies like the Uganda Institute of Professional Engineers (UIPE) have over other social organizations?**

MARKER WORDS: PROFESSIONAL BODY

Say it like this:

> A professional body regulates standards, certifies, licenses or registers practitioners, provides continuing professional development, advocates for and represents members, and enforces a code of ethics and discipline.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
regulation of standards	giving the first feature (in both solution files)	MUST
certification and licensing	explaining control of entry	MUST
continuing professional development (CPD)	explaining how members stay competent	MUST
advocacy and representation	explaining how the body speaks for members	MUST
ethics and discipline / disciplinary procedures	explaining enforcement of the code	MUST
professional body or association	describing who regulates a profession	MUST

▶ Recall drill: write the sentence from memory, then click each blank to check

FLASHCARDS: CLICK TO FLIP

The five distinguishing features of a professional body such as UIPE.
Regulation of standards; certification and licensing; continuing professional development (CPD); advocacy and representation; ethics and discipline.

The eight features that qualify engineering as a profession (short form).
Specialized knowledge; long formal education; registration and licensing; a professional body; a code of ethics; autonomy and judgment; public service and trust; accountability and discipline.

Factors that motivate engineers to practise their profession well

The ethics-cat-1 question says: "discuss the factors that would motivate engineers to practice their profession while carrying out their day-to-day activities". Read it as: what drives engineers to practise *ethically and*

competently every day, not only to turn up for work. Say this reading in your first line. It is a "discuss" question, so give four or five factors, each with a reason and an engineering example.

beyond the slides The factors, grouped:

- **Public safety and welfare.** Engineers know people use and live with what they build. The first canon, holding paramount the safety, health and welfare of the public, is a constant reason to practise carefully.
- **The code of ethics and professional standards.** The UIPE code and its canons tell the engineer what is expected. Following it gives clear direction in daily decisions.
- **Registration, licensing and discipline.** An engineer who breaks the code risks warnings, suspension and loss of registration. This is a strong external motive.
- **Legal liability.** Negligence can lead to lawsuits, compensation and criminal charges.
- **Personal integrity, conscience and professional pride.** Good engineers want to be able to stand behind their work. This is an internal, virtue-based motive.
- **Reputation and career progress.** Honest, competent work builds trust with clients, employers and peers, brings promotion, and wins future contracts.
- **Public trust in the profession.** Each engineer's conduct affects how society views all engineers.
- **Employer expectations and a good ethical climate.** Management that supports safety reporting and rewards honest work encourages ethical practice.
- **Fair rewards and working conditions.** Decent pay and recognition reduce the temptation to take shortcuts or accept bribes.
- **Continuing professional development.** Staying current keeps the engineer competent, which is itself an ethical duty.

Q: ethics-cat-1 Q1b · marks not stated **Discuss the factors that would motivate engineers to practice their profession while carrying out their day-to-day activities.**

FLASHCARDS: CLICK TO FLIP

Name five factors that motivate engineers to practise ethically (short form).
Public safety and welfare; code of ethics; registration and discipline; professional integrity and pride; reputation and public trust.

Whistleblowing: meaning, internal versus external

Whistleblowing is the act of an employee or member of an organisation disclosing information about wrongdoing in that organisation, such as a danger to public safety, illegal activity, fraud or corruption, to people who can act on it. The whistleblower is usually motivated by concern for the public and not by personal gain. beyond the slides A whistleblower acts when normal reporting has failed, and the act often goes against the employer's wishes. That conflict is why it is an ethical dilemma: loyalty to the employer against duty to the public.

The Set 4 model rules (slide wording). The Set 4 engineering code lists "Whistle blowing" under obligation to society. Rule 3 says: "If an engineer's professional judgment is overruled resulting in danger to the life, health, welfare, or property of the public, the engineer shall notify his/her employer or client and any appropriate authority." Rule 8, the "Final Obligation to Society", says engineers who know of a possible violation of the rules "shall provide pertinent information and assist the state board in reaching final determination of the possible violation". So the code does not ask the engineer to keep quiet: it requires notification first of the employer or client, then of the appropriate authority.

Internal versus external whistleblowing

Point	Internal whistleblowing	External whistleblowing
Who is told	Someone inside the organisation: supervisor, higher management, ethics officer, audit committee or board	Someone outside: regulator, police, a government authority, the professional body, the press, or the public
Aim	Let the organisation correct the problem itself	Force correction when the organisation will not act, or protect the public
Risk to whistleblower	Lower, but still possible retaliation	Higher: dismissal, legal action, loss of reputation
Risk to organisation	Contained, problem handled privately	Public exposure, fines, reputational damage
When used	First step, always tried first	Last resort, after internal channels have failed
Example	The BART engineers' memos to their managers (2017 paper Q4, see section t07)	Dr Hansen speaking to the press about NASA climate censorship

Q: **2013/14 Q4b and 2011/12 Q4b · 4 marks** **Distinguish between internal and external whistleblowing.**

MARKER WORDS: WHISTLEBLOWING MEANING AND TYPES

Say it like this:

> Whistleblowing is disclosing information about wrongdoing within an organisation that endangers the public; it is internal when reported within the organisation and external when reported to outsiders such as a regulator or the media.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
whistle blowing	naming the act (slide heading in Set 4, obligation to society)	MUST
notify employer or client and any appropriate authority	quoting the code on what to do when overruled (Set 4, rule 3)	SHOULD
danger to the life, health, welfare or property of the public	saying when whistleblowing is justified (Set 4)	SHOULD
internal channels	describing internal whistleblowing	SHOULD
last resort	explaining when external whistleblowing is allowed	SHOULD
duty to the public versus loyalty to the employer	naming the dilemma	SHOULD

▶ Recall drill: write the sentence from memory, then click each blank to check

FLASHCARDS: CLICK TO FLIP

Define whistleblowing.
Disclosing information about wrongdoing within an organisation, such as danger to the public, illegal acts or corruption, to people who can act on it.

Internal whistleblowing.
Reporting wrongdoing to someone within the organisation, through its internal channels.

External whistleblowing.
Disclosing wrongdoing to someone outside the organisation, such as a regulator, the police, the professional body or the media.

Set 4 rule 3 on whistle blowing (engineer overruled).
If an engineer's professional judgment is overruled resulting in danger to the life, health, welfare, or property of the public, the engineer shall notify his/her employer or client and any appropriate authority.

Conditions for blowing the whistle, critical loyalty and impacts

Conditions that must be met

beyond the slides The standard conditions come from the ethicist Richard De George. They answer the question: when is a person morally justified in blowing the whistle against an employer? Papers ask for three, so learn these three first and the extra two as a bonus.

1. **The harm is serious and considerable.** The wrongdoing must threaten serious harm to the public or to others, for example a defect that can kill. Trivial complaints do not justify going against the employer.
2. **The matter has been reported internally first.** The person has told the immediate superior and, if that failed, has used all available internal channels up to the highest authority. External whistleblowing comes only after these fail.
3. **The evidence is documented and convincing.** The person has good evidence that would persuade a reasonable, impartial observer that the harm is real. Rumour is not enough.
4. *Extra:* The person has good reason to believe that going public will actually prevent the harm. If it will achieve nothing, the cost to the whistleblower is hard to justify.
5. *Extra:* The motive is good faith concern for the public, not revenge, personal gain or spite.

Watch out. The question bank has the true-or-false statement "To qualify as a whistleblower, one must be an expert in the field in which they suspect illegal acts to be occurring." This is false. What is needed is reasonable belief supported by evidence, not expert status. A clerk who sees fraud can blow the whistle.

Q: 2013/14 Q4a and 2011/12 Q4a · 6 marks **Identify and explain three (3) essential conditions for one intending to "blow the whistle" against his or her employer.**

Critical loyalty

beyond the slides **Critical loyalty** is loyalty to an employer that is not blind obedience. The loyal employee is committed to the organisation's real, legitimate interests and long-term good, and therefore speaks up, questions decisions and reports wrongdoing when it would harm the organisation, its customers or the public. It contrasts with **uncritical or blind loyalty**, where the employee obeys every instruction and keeps quiet about wrongdoing. The key idea is that criticism through the right channels is a form of loyalty: a company that sells unsafe products will eventually suffer lawsuits, fines and loss of trust, so the engineer who reports the fault is protecting the employer. This is how the engineer can reconcile whistleblowing with the duty to act as a "faithful agent or trustee" of the employer (Set 4, model rules).

Q: **2013/14 Q4c and 2011/12 Q4c · 3 marks** **What do you understand by "critical loyalty" to an employer?**

Impacts on the whistleblower

The 2013/14 paper (Q4d; marks not legible in the scan, 7 by subtraction from the 20-mark total) asks:

EXAM WORDING

Discuss the most likely impacts on the whistle blower in an organization after they have revealed information.

The impacts are mostly negative for the person, though some are positive.

- **Retaliation and dismissal.** Sacking, demotion, transfer to a worse post, denial of promotion, or non-renewal of a contract.
- **Harassment and isolation.** Colleagues who fear for their own jobs may shun the person, and management may bully or monitor them.
- **Blacklisting.** Difficulty getting a new job in the same industry because employers see the person as a troublemaker. In Jenny's case (section t06) this is the same fear a whistleblower like Jenny faces.
- **Financial and legal strain.** Loss of income, legal costs, and possible lawsuits for breach of confidentiality or defamation.
- **Psychological and family stress.** Anxiety, depression, and strain on family relationships.
- **Positive effects.** Possible protection under whistleblower laws, support from the professional body, vindication when the wrong is proved, public respect, and reform of the organisation.

Q: **2013/14 Q4d · marks as printed, garbled** **Discuss the most likely impacts on the whistle blower in an organization after they have revealed information.**

Case: Dr James Hansen and NASA

On 29 January 2006 the New York Times reported that Dr James E. Hansen, a NASA climate scientist and Director of the Goddard Institute for Space Studies, said NASA headquarters officials had told public affairs staff to review his coming lectures, papers and website postings and his requests for interviews from journalists. He said the administration of President Bush was trying to stop him from speaking out since he gave a lecture calling for prompt reductions in emissions of greenhouse gases linked to global warming. Hansen said: "They feel their job is to censor information going out to the public." NASA's deputy assistant administrator for public affairs, Dean Acosta, replied that there was no effort to silence Hansen, that this was the usual way of operating at NASA, and that NASA promotes openness. The paper noted that Hansen had issued public warnings about the long-term threat of heat-trapping emissions since 1988, and that he had run-ins with appointees in various administrations. The paper asks you to identify Hansen's dilemma and say whether he acted unethically in insisting on speaking out.

Q: **2011/12 Q4(d) · [reconstructed from garbled scan] (i) 3 marks; (ii) marks not shown** **(i) Identify Dr Hansen's ethical dilemma. (ii) Did Hansen do something unethical by insisting on speaking out? Support your answer.**

For the full BART whistleblower question (2017 Q4D, **6 marks**:

EXAM WORDING

Use two different ethical theories to explain/justify the actions of a whistle-blower

), see section t07. For Marlinspike and surveillance work, see section t18.

MARKER WORDS: CONDITIONS, CRITICAL LOYALTY AND IMPACTS

Say it like this:

> A person may blow the whistle when the harm is **serious**, **internal channels have been exhausted** and there is **documented evidence**; **critical loyalty** means loyalty to the employer's legitimate interests without **blind obedience**.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
serious harm	giving the first condition	SHOULD
exhausted internal channels	giving the second condition	SHOULD
documented evidence	giving the third condition	SHOULD
critical loyalty	answering the 3-mark question EXAM WORDING What do you understand by 'critical loyalty' to an employer? (2011/12 and 2013/14 papers, Q4c)	MUST
blind obedience	contrasting with critical loyalty	SHOULD
retaliation	describing impacts on the whistleblower	SHOULD
faithful agent or trustee	linking whistleblowing to loyalty (Set 4 and the solution files)	MUST
dismissal, demotion, blacklisting	listing the personal costs	SHOULD

▶ Recall drill: write the sentence from memory, then click each blank to check

FLASHCARDS: CLICK TO FLIP

The three essential conditions for whistleblowing.
The harm is serious and considerable; internal channels have been used and exhausted; there is documented evidence that would convince a reasonable, impartial observer.

Define critical loyalty.
Loyalty to the employer's legitimate interests that is not blind obedience: the employee questions and reports wrongdoing, through proper channels, when it would harm the organisation or the public.

Four likely impacts on a whistleblower.
Retaliation (dismissal or demotion); harassment and isolation; blacklisting and financial hardship; psychological and family stress.

Set 4 rule 8, the Final Obligation to Society.
Engineers who have knowledge of possible violation of any of the rules shall provide pertinent information and assist the state board in reaching final determination of the possible violation.

Conflict of interest: meaning, actual versus potential

A **conflict of interest** is a situation in which a person's private interest, such as financial gain, a family tie, friendship or another loyalty, conflicts with, or could appear to conflict with, the professional duty to act in the best interest of the employer, client or public, so that the person's professional judgment may be compromised. The key idea is that the engineer must act as a faithful agent or trustee, and a competing interest threatens objectivity. Note that a conflict of interest is a *situation*, not an act. The wrong comes when it is hidden or when the engineer acts on the private interest.

Actual versus potential

Type	Meaning	Example
Actual conflict of interest	The conflict exists now: the person has the competing interest and is in a position to decide or influence a matter, so judgment is in fact affected or is being tested.	An engineer on a tender panel evaluates a bid from a company owned by her brother.
Potential conflict of interest	The competing interest exists, but no decision has yet been affected; the situation could become an actual conflict in future.	An engineer holds shares in a supplier that the company might use next year, and no purchase is pending.

beyond the slides Some textbooks add an **apparent** (or perceived) conflict: an outsider could reasonably think the judgment is compromised even if it is not. Set 4 clause 2 covers all three: disclose "potential conflicts of interest or any other circumstances that could influence or appear to influence" professional judgment.

The Set 4 model rules (slide wording)

The Set 4 slides give these conflict-of-interest clauses, grouped by who the obligation is owed to. Learn them as quotations.

- **Employers and clients, rule 1.** "Engineers shall not solicit or accept direct or indirect considerations, financial or otherwise, from contractors, their agents, or other parties while performing work for employers or clients."
- **Employers and clients, rule 2.** "Engineers shall disclose to their employers or clients potential conflicts of interest or any other circumstances that could influence or appear to influence their professional judgment or their service quality."
- **Full disclosure.** "An engineer shall not accept financial or other compensation from more than one party for services rendered on one project unless the details are fully disclosed and agreed by all parties."
- **Government conflicts of interest.** "To avoid conflicts of interest, engineers shall not solicit or accept a professional contract from a governmental body on which a principal or officer of their firm serves as a member. An engineer who is a principal or employee of a private firm and who serves as a member of a governmental body shall not participate in decisions relating to the professional services solicited or provided by the firm to the governmental body."
- **Obligations to other engineers, conflicts of interest.** "Engineers shall not directly or indirectly give, solicit, or receive any gift or commission, or other valuable consideration, in order to obtain work, and shall not make contribution to any political body with intent of influencing the award of contract by governmental body."
- **Obligation to society, the duty of full disclosure.** Engineers shall not express a professional opinion on matters for which they are motivated or paid unless they identify the parties on whose behalf they speak and reveal those parties' interest.
- **Canon:** act for each employer or client as *faithful agents or trustees*, and avoid conflicts of interest. See section t09 for the full list of obligations and section t11 for the canons.

How to handle a conflict of interest

beyond the slides Four steps: (1) **identify** the conflict early; (2) **disclose** it in writing to the employer, client or committee; (3) **withdraw or recuse** yourself from the decision, or divest the competing interest; (4) **let someone independent decide** and keep a record. Disclosure alone is not always enough: if the interest is large, you must step aside.

Q: 2013/14 Q5a · 2 marks; 2011/12 Q5a · 2 marks **What do you understand by "conflict of interest"?**

Q: 2013/14 Q5b, 2011/12 Q5b, CAT 2 2013/14 Q1a, CAT 2 2014/15 Q1a · 2 to 4 marks **Distinguish between actual and potential conflicts of interest.**

Q: CAT 2 2014/15 Q1b and CAT 2 2013/14 Q1b · 2 to 4 marks **Describe two unique situations in which an employee may experience an actual conflict of interest.**

Q: 2013/14 Q5c and 2011/12 Q5c · 6 marks **Identify three (3) unique situations that can present conflict of interest for an engineer.**

MARKER WORDS: CONFLICT OF INTEREST, ACTUAL AND POTENTIAL

Say it like this:

> A conflict of interest is a situation where a private interest conflicts with professional duty and may compromise professional judgment; it is actual when it exists now and potential when it could arise, and it must be disclosed to the employer or client.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
conflict of interest	naming the issue (Set 4 slide, papers)	MUST
actual / potential conflict of interest	distinguishing the two (papers: Set 4 mentions only "potential")	MUST
influence or appear to influence professional judgment	defining the harm (Set 4 rule 2)	SHOULD
disclose to employers or clients	saying what to do (Set 4 rule 2)	SHOULD
faithful agent or trustee	naming the duty that is threatened	MUST
full disclosure	explaining dual payment (Set 4)	SHOULD
direct or indirect considerations (gifts, commissions)	describing bribery situations (Set 4)	SHOULD
government conflicts of interest	explaining the part-time town engineer	SHOULD

▶ Recall drill: write the sentence from memory, then click each blank to check

FLASHCARDS: CLICK TO FLIP

Define conflict of interest.
A situation in which a person's private interest conflicts with their professional duty, so that their professional judgment may be (or appear to be) compromised.

Actual conflict of interest.
A conflict that exists now: the person has the competing interest and is in a position to influence the decision.

Potential conflict of interest.
A competing interest that has not yet affected a decision but could become an actual conflict in the future.

Set 4 rule 2 on conflict of interest (employers and clients).
Engineers shall disclose to their employers or clients potential conflicts of interest or any other circumstances that could influence or appear to influence their professional judgment or their service quality.

Set 4 rule on full disclosure (double compensation).
An engineer shall not accept financial or other compensation from more than one party for services rendered on one project unless the details are fully disclosed and agreed by all parties.

Three situations that present conflict of interest.
Accepting gifts or commissions from contractors; financial or family interest in a bidder or supplier; dual roles or double payment on one project.

Conflict of interest cases: Engineer A and B, and Maggie and Red Sky

Case 1: Engineer A, Engineer B and Smithtown

Engineer A is the part-time town engineer for Smithtown and also runs a consulting engineering practice. With A's advice and agreement, the town hires Engineer B to design a local road project. After B starts the preliminary design, A, acting as town engineer, reviews B's work and becomes convinced that B's performance does not meet the standards in B's contract with the town. B is terminated under the terms of his contract. A then offers, and Smithtown agrees, that A's own firm will do the road design. The questions: was it ethical for A to tell the town that B's performance fell short? Was it ethical for A to offer and agree to do the design work?

Q: **2013/14 Q5(d) · 8 marks** **Engineer A is the part-time town engineer for Smithtown and also has a consulting engineering practice. [Case as above.] 1. Was it ethical for Engineer A to contact Smithtown and advise the town that Engineer B's performance did not meet the standards outlined in the contract? 2. Was it ethical for Engineer A to offer and agree to perform the road design work for Smithtown?**

Case 2: Maggie and Red Sky

The scan of this question is badly garbled. Reconstructed: Maggie works for Harold Telecom and has been placed in charge of finding a firm to design and produce promotional posters and billboard ads for the launch of a new product line in two months. Red Sky, an advertising and marketing company, is one of the firms expected to bid. Maggie heads the team that will select the most suitable firm. The sentence that states Maggie's link to Red Sky is lost in the scan, but the question is clearly a conflict-of-interest case, so the answer assumes she has a personal tie to Red Sky (a relative, friend or financial interest). Part (i) asks her ethical dilemma (4 marks), part (ii) what she should do (4 marks).

> **Q:** **2011/12 Q5(d) · 4 + 4 marks · [reconstructed from garbled scan]** **Maggie is in charge of the search for a firm to design and produce promotional posters and billboard ads when Harold Telecom launches a new line of products in two months. Red Sky is one of the companies expected to bid. As head of the team selecting the most suitable firm: (i) Identify Maggie's ethical dilemma. (ii) What do you suggest Maggie should do?**

The Phillip case (an employee reading a client's e-mails, CAT 2 2013/14 Q1c:

> EXAM WORDING
> Describe Phillip's ethical dilemma

(**3 marks**) and

> EXAM WORDING
> What should Phillip as young professional do in these circumstances?

(**3 marks**)) and Jenny's case (CAT 2 2014/15 Q2, **12 marks**:

> EXAM WORDING
> Describe Jenny's ethical dilemma and explain what should she do as a young professional in her circumstances?

) are dilemma-resolution cases about reporting wrongdoing; they belong to section t06. They use the same idea as this section: weigh duty to the employer against duty to the public, and report through proper channels first.

MARKER WORDS: CONFLICT OF INTEREST CASES

Say it like this:

> In a conflict of interest case, identify the competing interest, say whether it is actual or potential, then disclose it, withdraw from the decision and let an independent party decide.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
faithful agent or trustee	naming Engineer A's or Maggie's duty to the client	MUST
actual conflict of interest	classifying Engineer A's second step	SHOULD
potential conflict of interest	classifying Maggie before the selection	MUST
disclose	giving the first remedy	SHOULD
recuse / withdraw from the decision	giving the second remedy	SHOULD
appear to influence professional judgment	explaining why appearance matters (Set 4 rule 2)	SHOULD
government conflicts of interest	applying the town-engineer clause (Set 4)	SHOULD

▶ Recall drill: write the sentence from memory, then click each blank to check

FLASHCARDS: CLICK TO FLIP

Four steps for handling a conflict of interest.
Identify it early; disclose it in writing; withdraw or recuse from the decision (or divest the interest); let someone independent decide and keep a record.

Set 4 government conflicts of interest rule (second sentence).
An engineer who is a principal or employee of a private firm and who serves as a member of a governmental body shall not participate in decisions relating to the professional services solicited or provided by the firm to the governmental body.

Ethics of the internet, AI, robots and the environment

~35 min to read | High priority | 10 questions

Source: not in your slides. Built from the question pools (ethics-cat-1 Q3c, 2017 Q2 A-B, CAT 2021-22 Q1.2, QB C3, 2011-2015 papers) and general knowledge. Everything here is **beyond the slides**, so the wording is a safe model, not the lecturer's own.

> **About tags in this section.** Everything here is beyond the slides. A Must tag means the phrase appears in the past papers or solutions, not in the slides.

> **How these questions are marked.** They ask for an issue and then "the ethical dilemma therein". The 2017 paper (Q2B, **9 marks**:
>
> EXAM WORDING
>
> Describe three ethical issues that arise from internet and usage. For each issue highlight the ethical dilemma therein i.e. benefits vs harm, legality vs morality
>
>) spells out what a dilemma means here: **benefits vs harm**, **legality vs morality**. So for every issue, write one line in the form "the benefit is X, but the harm is Y, so the dilemma is whether...". Use that pattern throughout. For the general idea of an ethical dilemma and how to resolve one, see Law versus ethics and resolving ethical dilemmas.

Ethical issues of AI in education

AI in education means tools such as chatbots and writing assistants, automatic marking, adaptive learning platforms, proctoring software and student-analytics systems. They save time and widen access. They also create ethical problems. Each issue below is written as: the issue, why it arises, and the dilemma as benefit versus harm.

1. Academic dishonesty and plagiarism

Students can generate essays, code and answers with AI and submit them as their own. This is a form of cheating and passes off someone else's work as yours. It also makes it hard for lecturers to know what a student really knows. Detectors are unreliable and sometimes accuse honest students. For the general rules on plagiarism and cheating see Academic integrity.

Dilemma: AI helps students learn and work faster (benefit), but it lets them submit work that is not theirs and weakens honest assessment (harm). Should it be allowed as a study aid, or banned to protect integrity?

2. Privacy and data protection

AI tools collect student data: essays, grades, behaviour, voice, faces and keystrokes. Companies may store it, profile students or sell it. Students, and especially minors, often cannot give informed consent. Data can be breached or used for purposes never agreed. See profiling and monitoring for the privacy material.

Dilemma: More data lets AI personalise learning and spot struggling students early (benefit), but it invades privacy and exposes students to misuse and breaches (harm).

3. Bias and unfairness (algorithmic bias)

AI learns from past data. If the data under-represents some groups, accents, languages or regions, the system marks or recommends unfairly. An essay scorer may penalise non-native English. A system trained mostly on foreign data may not fit African students. Bias is hidden because the system looks objective.

Dilemma: Automated decisions are fast, consistent and cheap (benefit), but they may discriminate against some students without anyone noticing (harm). Is a consistent unfair mark better than an inconsistent human one?

4. Digital divide and unequal access

AI tools need devices, electricity, data bundles and good internet. Students in rural areas or poor families cannot use them, while others gain an advantage. Paid versions are better than free ones. This widens existing inequality.

Dilemma: AI can democratise learning for those who can reach it (benefit), but it deepens the gap between privileged and marginalised students (harm). Should institutions adopt it at all before access is equal?

5. Over-reliance and loss of critical thinking

If students let AI think, write and solve for them, they may not build their own skills: reasoning, writing, problem solving and creativity. Graduates then lack competence, which matters most in engineering where unsafe work harms the public. Teachers may also over-trust AI feedback.

Dilemma: Convenience and quick help (benefit) against weaker learning, independence and real competence (harm).

6. Accuracy, transparency and accountability

AI can produce confident but false answers (hallucinations), fake references and outdated content. Most systems are "black boxes", so nobody can explain why a grade or recommendation was given. When a student is wrongly failed or flagged, it is unclear who is responsible: the developer, the school or the teacher.

Dilemma: Efficiency and instant answers (benefit) against errors that cannot be explained or challenged, with no one accountable (harm).

7. Surveillance and loss of autonomy (proctoring)

Exam proctoring AI watches faces, eyes, sounds and rooms. It can mistake normal behaviour for cheating, and it intrudes into the home. Constant monitoring creates stress and removes students' freedom.

Dilemma: Preventing cheating protects fairness (benefit), but intrusive monitoring violates privacy and dignity (harm). This is a privacy versus security trade-off, as in privacy and government surveillance.

8. Impact on teachers, human contact and jobs

Heavy use of AI may replace teaching and mentoring, reduce human care and relationships, and threaten teachers' jobs. Education also builds values, which a system cannot model well.

Dilemma: Cost saving and wider reach (benefit) against loss of human guidance, empathy and employment (harm).

9. Intellectual property and consent

AI is trained on writing and images of others, often without permission. Students may submit AI output that copies protected work. See intellectual property.

Dilemma: Free access to knowledge and creativity (benefit) against the rights of the original creators to credit and reward (harm). It is also a legality versus morality question where laws lag behind the technology.

> **Exam tip (6 marks).** Six marks means three issues with a dilemma each (about one mark for the issue and one for the dilemma), but give four if you have time. Choose the easiest: dishonesty, privacy, bias, digital divide. Always use the words "on one hand... on the other hand", "benefit" and "harm".

> **Q:** `ethics-cat-1 Q3c · 6 marks` **Describe ethical issues that arise from the use of technologies like artificial intelligence for educational purposes. For each issue, highlight the ethical dilemma therein.**

MARKER WORDS: AI IN EDUCATION

Say it like this:

> AI in education raises ethical issues of academic dishonesty, privacy, bias, digital divide, over-reliance and accountability; in each, the dilemma is the benefit it brings against the harm it can cause.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
benefit vs harm	stating any dilemma (the 2017 paper uses this wording)	MUST
privacy, consent	discussing student data	MUST
bias, fairness	discussing marking or ranking by AI	MUST
accountability, transparency	asking who is responsible for AI errors	MUST
plagiarism, academic integrity	discussing cheating with AI	MUST
digital divide	discussing unequal access	MUST
over-reliance, critical thinking	discussing skill loss	SHOULD
black box, hallucination	discussing accuracy and explainability	SHOULD

▶ Recall drill: write the one-line answer from memory, then click each blank to check

FLASHCARDS: CLICK TO FLIP

Six ethical issues of AI in education.
Academic dishonesty and plagiarism; privacy and data protection; bias and unfairness; digital divide; over-reliance and loss of critical thinking; accuracy, transparency and accountability.

What does "ethical dilemma" mean in these questions?
A choice where a benefit comes with a harm (benefit vs harm), or where what is legal may not be moral (legality vs morality).

Algorithmic bias.
Unfair results from an AI system because its training data under-represents or misrepresents some groups.

The internet: benefits and ethical issues

Benefits

The question asks for four or five benefits "that have made our lives better". Name each, then add a short example.

- **Access to information.** A vast store of knowledge on almost any topic, for research and self-learning.
- **Communication and connectivity.** Real-time contact across the world by email, social media, video calls and messaging. Families, teams and businesses stay linked.
- **Convenience and efficiency.** Online banking, mobile money, shopping, travel booking and government services save time and cost.
- **Education and skill development.** Online courses and tutorials open learning to people who cannot attend a campus.
- **Commerce and jobs.** E-commerce, remote work and online marketing create business and income. beyond the slides
- **Entertainment and civic participation.** Media, news and public debate; citizens can hold leaders to account. beyond the slides

Ethical issues, each with its dilemma

Issue	What goes wrong	Dilemma (benefit vs harm, legality vs morality)
Privacy concerns	Companies and governments collect and analyse personal data for targeted adverts and surveillance, often without clear consent.	The right to privacy against the need for data collection and surveillance for security or commercial purposes.
Cybersecurity threats	Hacking, malware, phishing and data breaches. The same techniques can be used for espionage or financial gain.	Pursuing technological advancement against the responsibility to protect people and organisations from attack. Is it ethical to weaponise technology?
Digital divide	Unequal access to internet infrastructure, devices and digital skills between rich and poor, urban and rural.	Spreading the internet's benefits to all against the cost, which leaves marginalised groups with fewer chances in education, jobs and civic life.
Intellectual property and piracy	Easy copying and sharing of music, software and books online. See protection challenges online	Freedom of access to information against the creator's rights. Legal in one country may be illegal in another.
Harmful content and misinformation	Fake news, cyberbullying, hate speech and pornography spread quickly and anonymously.	Freedom of expression against protecting people from harm. Who decides what is censored?

The first three rows are the standard model answer (CAT 2021-22 Q1.2, marks not printed:

EXAM WORDING

Describe three ethical issues that arise from internet access and usage

). The last two are alternates if you prefer them. For the privacy-versus-security discussion on monitoring and censoring the internet, see privacy and government surveillance. The smart-city privacy case is handled there too and is not repeated here.

Q: **2017 Q2 A · 5 marks** **Internet data share and applications have become very much an integral part of our human day to day lives. List 5 benefits facilitated by the internet that have made our lives better.**

Q: **2017 Q2 B · 9 marks (same as CAT 2021-22 Q1.2 II)** **Describe three ethical issues that arise from internet access and usage. For each issue highlight the ethical dilemma therein i.e. benefits vs harm, legality vs morality.**

MARKER WORDS: INTERNET BENEFITS AND ISSUES

Say it like this:

> The internet gives access to information, communication and connectivity, convenience and efficiency and education, but raises privacy concerns, cybersecurity threats and a digital divide, each a dilemma of benefits vs harm.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
access to information	listing benefits	MUST
communication and connectivity	listing benefits	MUST
convenience and efficiency	listing benefits	MUST
privacy concerns	first ethical issue	MUST
cybersecurity threats	second ethical issue	MUST
digital divide	third ethical issue	MUST
legality vs morality	naming the dilemma type	SHOULD
socioeconomic inequalities	explaining the digital divide	MUST

▶ Recall drill: write the answer from memory, then click each blank to check

FLASHCARDS: CLICK TO FLIP

Four benefits of the internet (CAT 2021-22 model answer).
Access to information; communication and connectivity; convenience and efficiency; education and skill development.

Three ethical issues of internet use (model answer).
Privacy concerns; cybersecurity threats; digital divide.

Digital divide.
The gap in access to internet infrastructure, devices and digital skills, which worsens socioeconomic inequality.

Robots and the ethical values to embed

Robots and autonomous systems (self-driving vehicles, industrial robots, care robots, drones) act in the physical world. A fault or a bad decision can injure or kill. They cannot be held morally responsible themselves, so the designers must build in values in advance so that the machine behaves ethically when no human is watching.

Values that should be embedded

- **Safety and protection of human life.** The first rule: the robot must not harm people, and must stop or choose the least harmful action when danger arises. This echoes the first canon of engineering codes, to hold paramount the safety of the public. See codes of ethics.
- **Respect for human dignity and rights.** Including privacy, autonomy and non-discrimination.
- **Fairness and non-bias.** Decisions must not favour or harm people because of race, sex, age or status.
- **Transparency and explainability.** People must be able to understand why the robot acted as it did.
- **Accountability and responsibility.** A human, the designer, manufacturer or operator, stays answerable for the robot's actions, and decisions can be traced and logged.
- **Honesty and trust.** Robots should not deceive users about what they are or what they can do.
- **Beneficence and non-maleficence.** Do good, avoid harm; the benefit must exceed the risk.
- **Human oversight and control.** A human can override or shut down the robot. This is sometimes called "meaningful human control".

A classic reference is Asimov's three laws: a robot may not injure a human; it must obey humans unless that conflicts with the first law; it must protect itself unless that conflicts with the first two. beyond the slides The laws are a starting point, but real situations need richer values and ethical theories: a utilitarian robot weighs total harm, a deontological one follows fixed rules (see ethical theories). A self-driving car forced to choose whom to hit shows the dilemma.

Q: QB C3 · 5 marks **Robots raise several ethical issues especially concerning safety of human lives. Which ethical values should have been embedded in them to enable them to make ethical decisions?**

MARKER WORDS: ETHICAL VALUES FOR ROBOTS

Say it like this:

> Robots should be embedded with **safety of human life**, **human dignity**, **fairness**, **transparency** and **accountability**, with **human oversight**.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
safety of human life	giving the first and main value	SHOULD
accountability	saying who answers for a robot's act	MUST
transparency	asking for explainable decisions	MUST
fairness	avoiding bias	MUST
human dignity	respect for rights	SHOULD
human oversight	override and shutdown	SHOULD
non-maleficence	do no harm	SHOULD

▶ Recall drill: write the sentence from memory, then click each blank to check

FLASHCARDS: CLICK TO FLIP

Ethical values to embed in robots (exam list).
Safety of human life; respect for human dignity and rights; fairness; transparency; accountability; human oversight.

Asimov's three laws.
1) A robot may not injure a human. 2) It must obey humans unless that conflicts with law 1. 3) It must protect itself unless that conflicts with laws 1 and 2.

Environment and sustainable development (compact block)

This is a fixed set of questions that repeats across the 2011 to 2015 papers. Exact wording (2014/15 Q3, with its marks): (a)

EXAM WORDING

What do you understand by 'Sustainable development'?

(**2 marks**); (b)

> **EXAM WORDING**
> Name two forms of environmental pollution and explain one problem associated with each

(**4 marks**); (c)

> **EXAM WORDING**
> Identify and explain three principles that guide engineers in making environmental ethics decisions

(**6 marks**); (d) (i)

> **EXAM WORDING**
> Identify two ethical problems associated with increasing the use of fossil fuel as a source of energy in the manufacturing and transport industries

(**4 marks**) and (ii)

> **EXAM WORDING**
> Suggest one ethically sound solution to each of these problems

(**4 marks**). For the nuclear-power and Fukushima reasoning, and the argument about depleting resources, which are in your slides, see why ethics belongs in engineering.

Sustainable development

Sustainable development is **development that meets the needs of the present without compromising the ability of future generations to meet their own needs**. It balances economic growth, social well-being and environmental protection. **beyond the slides** (Brundtland definition)

Forms of pollution and one problem each

Form	A problem associated with it
Air pollution	Smoke, vehicle fumes and industrial gases cause respiratory disease and contribute to acid rain and global warming.
Water pollution	Industrial effluent, sewage and chemicals poison drinking water and kill fish, causing disease and loss of livelihoods.
Land or soil pollution	Dumping, plastic and e-waste contaminate soil and reduce farmland fertility, harming food security.
Noise pollution	Machinery and traffic cause stress, hearing loss and sleep disturbance.
Radioactive pollution	Nuclear leaks and waste cause cancer and contaminate land for generations.

Environmental ethics principles for engineers

- **Sustainability / intergenerational justice.** Do not deplete resources or damage the environment in ways that deny future generations a fair chance.
- **Precautionary principle.** Where an action may cause serious or irreversible damage, lack of full scientific proof is not a reason to delay protective measures.
- **Polluter pays / responsibility.** Those who cause pollution should bear the cost of preventing and cleaning it.
- **Stewardship and respect for nature.** Humans have a duty to care for the environment and its living things, which have value beyond their use to us.
- **Public safety, health and welfare first.** The canons of engineering codes require engineers to hold public safety paramount, which includes environmental safety. See codes of ethics.
- **Minimise harm and waste.** Use cleaner production, reuse and recycle, and design for reduced environmental impact. The 2013/14 paper (Q3a, **3 marks**:

  > EXAM WORDING
  > What do you understand by the phrase eco-redesign of a product?

) also asks about eco-redesign: redesigning a product to reduce its environmental impact over its whole life.

Fossil fuels: ethical problems and solutions

Ethical problem	Ethically sound solution
Climate change and air pollution: burning coal, oil and gas releases carbon dioxide and toxic gases that harm health and the climate, and the poorest suffer most.	Shift to **renewable energy** (solar, hydro, wind), improve efficiency and use cleaner technology such as filters and carbon capture.
Injustice to future generations non-renewable fuels are used up, and the damage is left to those not yet born.	Conserve and diversify energy sources; set long-term energy plans and invest in alternatives.
Harm to public health and communities: transport and factory emissions make people ill without their consent.	Enforce emission standards and the polluter-pays rule; promote public transport and electric vehicles.

Give two problems and one solution for each, as asked. A good solution answers its problem directly.

Q: **2011/12, 2013/14, 2014/15 papers · 2 marks** **What do you understand by 'Sustainable development'?**

Q: **2011-2015 papers · 4 marks** **Name two (2) forms of environmental pollution and explain one (1) problem associated with each.**

Q: **2011-2015 papers · 6 marks** Identify and explain three (3) principles that guide engineers in making environmental ethics decisions.

Q: **2011-2015 papers · 4 + 4 marks** (i) Identify two (2) ethical problems associated with increasing the use of fossil fuel as a source of energy in the manufacturing and transport industries. (ii) Suggest one (1) ethically sound solution to each of these problems.

Q: **2013/14 paper Q8(a) · 3 marks** What do you understand by the phrase "eco-redesign of a product"? [reconstructed from garbled scan; the scan reads "Fm-redesign ofa produce"]

Q: **2013/14 paper Q8(b) · 4 marks** Mention any two ways in which a telecommunications/electrical engineer can contribute towards sustainable development in a country. [reconstructed from garbled scan]

MARKER WORDS: ENVIRONMENT AND SUSTAINABLE DEVELOPMENT

Say it like this:

> Sustainable development is development that meets the needs of the present without compromising the ability of future generations to meet their own needs.

Vocabulary to weave into explanations:

Marker	Use it when you are...	Tag
sustainable development	defining or opening any environment answer	MUST
air pollution water pollution	naming forms of pollution	SHOULD
future generations	justifying sustainability or fossil-fuel limits	SHOULD
precautionary principle	stating a guiding principle	SHOULD
polluter pays	stating a guiding principle	SHOULD
renewable energy	giving a fossil fuel solution	SHOULD
public safety, health and welfare	linking to the code of ethics	SHOULD

▶ Recall drill: write the definition from memory, then click each blank to check

FLASHCARDS: CLICK TO FLIP

Sustainable development.
Development that meets the needs of the present without compromising the ability of future generations to meet their own needs.

Forms of pollution.
Air; water; land or soil; noise; radioactive.

Three environmental ethics principles for engineers.
Sustainability (intergenerational justice); precautionary principle; public safety, health and welfare.
Alternates: polluter pays; stewardship.

Precautionary principle.
Where an action may cause serious or irreversible harm, lack of full scientific proof is not a reason to delay protective measures.

APPENDIX

Question index

Every question in this guide, with where it sits. 180 questions.

What ethics is, Kantianism and utilitarianism

- 2017 Q1 · 2 marks each Give clear definitions of the following terms as they relate to the course Technology, Ethics and Human Rights: (i) Ethics (ii)...
- 2016 Q1a, b · CAT 2021-22 Q1.1, 1.2 Differentiate between the following: (a) Values and morals. (b) Ethics and morality.
- 2014/15 Q4a · 2011/12 Q1a · 2 marks each, 8 marks Define the following terms: (i) Virtue (ii) Conscience (iii) Ethical dilemma (iv) Intrinsic good.
- CAT 1 2015 Q1a-c · 5, 2 and 3 marks (scan order unclear) (a) Of what value is the study of the human person in Ethics? (b) Distinguish between human acts and acts of man. (c) Of what v...
- Practice · 4 marks Some people say that studying ethics is pointless because right and wrong are personal opinions. Give two arguments against thi...
- QB E2 · 6 marks · 2016 Q1d · CAT 2021-22 Q1.4 Characterize and distinguish between consequentialist ethical theories and deontological ethical theories. Give one example of ...
- 2017 Q1 · 4, 4, 4, 5 marks Define: Utilitarianism (4 marks). Deontology (4 marks). Teleology (4 marks). Consequentialist theory (5 marks).
- QB E1 · 6 marks What is the defining characteristic of consequentialist theories in ethics? Identify at least two problems with consequentialis...

- QB E4 · 6 marks · 2018 Morality c "Kant's idea of universalization does not work in practice." Discuss.
- 2018 Morality c How relevant is Kant's universalizability principle today?
- 2016 Q5A · 4 marks In your own words describe the central tenet of utilitarianism.
- 2016 Q5B · 6 marks · 2021-22 exam Q1d Describe THREE frequently cited criticisms of utilitarian ethics. [Two drawbacks in the 2021-22 version.]
- 2016 Q4A · 8 marks · CAT 2021-22 Q3b · [reconstructed from garbled scan] Construct two utilitarian-style arguments supporting, and two opposing, strong property rights over computer software.
- 2018 Morality a, b (a) Define utilitarianism and absolutism and discuss the failings of each principle. (b) Utilitarianism is largely driven by th...
- 2021-22 exam Q2A · 2017 Q4A Describe the different theories associated with morality (deontological, consequentialist and virtue ethics).
- 2021-22 exam Q4 A-D (A) Suppose James the next day decides to follow Bill's order, finishing the report at once and abandoning the other tests. Can...

Relativism, egoism, social contract and divine command

- 2016 Q1B · 8 marks Compare and contrast ethical egoism and duty ethics as decision-making frameworks for engineers.
- 2021-22 exam, Q2C · define with an example With the aid of examples define the following ethical concepts: ethical pluralism; ethical egoism; ethical relativism; ethical ...
- Set 1 slide 36-37 · discussion Which of these should be considered legitimate positive rights by our society: a minimum standard of living; housing; health ca...
- 2016 Q5C · 10 marks Identify the ethical theory that underlies each of the following 5 arguments for or against unauthorized online file-sharing of...
- Set 1 slide 35 What are some examples of contemporary technology issues for which our society's moral guidelines seem to be nonexistent or unc...
- Set 1 slide 35 (Ethical vs legal) Can you give examples where an action may be legal but unethical, or vice versa?
- Set 1 slide 35 Do (or should) organizations have ethics? If so, who makes them? And who are affected: employees, customers, the society?
- Set 1 slide 36 Which of the following should be considered legitimate positive rights by our society: minimum standard of living, housing, hea...
- Set 1 slide 37 Give examples of conflicts between the positive rights of one person and the negative rights of another.

Why ethics belongs in engineering

- 2011 / 2013 / 2015 papers (also QB, reversed wording) · 12 marks "Ethics has no place in the practice of the engineering profession." Discuss this statement.

- 2017 Q3E · 5 marks Explain the need for ethics in the engineering profession.
- Set 2 slide 3-4 clicker · practice A person's behaviour is always ethical when one: A. does what is best for oneself; B. has good intentions, no matter how things...

The engineering code of ethics and its cases

- CAT 1 2015 Q2a · 1 mark (also 2015 exam Q1a, 3 marks) State the principle of individual competence of an engineer.
- 2016 Q3C · 9 marks In March 1995, the structural engineer for the Olympic Stadium in Atlanta, Georgia, discovered an error in his design of a stee...
- 2015 Q3d · 7 marks (also 2011/12 paper) Two engineers submitted sealed bids to a prospective client for an engineering project. The client informed Engineer A of the a...
- 2015 Q1c · 8 marks (also 2011/12 paper) Distinguish between personal responsibility and organizational responsibility as far as morality is concerned.

Academic integrity and ethical conduct of research

- Slides 49-56 · exam-style · 5 marks Name and describe five types of cheating that a student can commit.
- Set 3 slide 2 · discussion You must choose between an applicant with almost all A grades from a university with reported cheating scandals, and an applica...
- Past papers 2011-2015 · 6 marks Identify and explain three (3) practices that violate ethical conduct of research.
- Past papers 2015 / 2013-14 / 2011-12 · 2 marks (d)(i) Describe Mildred's ethical dilemma.
- Past papers 2015 / 2013-14 / 2011-12 · 3 marks (d)(ii) Is Mildred's final project report a product of ethical conduct of research? Justify.

Law versus ethics, and resolving ethical dilemmas

- QB E3 · 7 marks What is the difference between ethics and law?
- Set 2 clicker Which of the following ensure that behaviour is ethical? I. Following the law. II. Acting in the best interest of society. III....
- 2017 Q3D · 6 marks (also 2021-22 exam Q2B) Explain the difference between moral and legal responsibility (mention two points of difference). Illustrate your explanation w...
- Set 4 slide 6 · case You are the manager at Big-Mart. You fired Al, a sales clerk, after he punched a customer. A Mega-Mart manager phones and asks ...
- CAT 2021-22 Q2d (also 2020-21 paper) Briefly describe how ethical dilemmas arise in the practice of the Engineering profession.
- Set 3 slides 6-7 · discussion A manufacturer prints in the product literature: "This product may contain unexpected flaws and might not operate correctly und...
- Set 3 slide 8 · Pentium Was there really an ethical dilemma in the Pentium case? If so, what was it?

- ethics-cat-1 Q3a · 2 marks; also asked as a 2-mark "define" in the 2011/12 and 2014/15 papers Explain with the aid of examples what you understand by the term ethical dilemma.
- ethics-cat-1 Q3b · 6 marks Briefly describe the steps a given professional would take while attempting to resolve ethical dilemmas.
- CAT 2 2015 · 12 marks Jenny recently completed her master's degree and was hired for her dream job with the local county government. In her first yea...
- CAT 2 2014 Q1c · 3 + 3 marks Phillip, a trainee telecommunications engineer working for a local Internet Service Provider's Technical Department, one mornin...

Case analysis and the 7-step template

- Set 3 case 1 · template practice Your employer asks you to design a bridge costing no more than $1 million. A $1m bridge collapses in a moderate earthquake; a $...
- Set 3 case 2 · template practice You are asked by the government to verify that a nuclear reactor will not leak toxic substances into the ocean. You find it wil...
- QB A1(i) · 12 marks Simon is working as a Social Media Consultant for a multi-national corporation. One day he gets a call from the CIO requesting ...
- QB C2 · 12 marks Aggie's pride and joy is her garden. One day she notices weeds growing through the gaps between the pavers in her garden. She d...
- 2017 Q4 · template walk-through Analyse the BART case using the seven-step template.
- 2017 Q4A · 6 marks What would you advise Holger Hjortsvang, Max Blankenzee and Robert Bruder to do after the response by the managers? Apply two o...
- 2017 Q4B · 4 marks If the two theories come to different answers, explain why you think the outcome of one theory should prevail over the outcome ...
- 2017 Q4D · 6 marks Use two different ethical theories to explain or justify the actions of a whistle-blower.
- 2015 Q6 · 20 marks The heart pacemaker is a modern wonder. The device has a timer that resets itself every time the patient's heart beats. If the ...

The four group cases

- Practice · 12 marks Workers at a plant remove machine guards to work faster, and they say they are happy to take the risk. As the new production en...
- Practice · 8 marks A colleague who handles effluent discharge tells you that exceeding the legal limit by a small margin is "not worth correcting"...
- Practice · 8 marks Your company wants to change the material of a product without telling customers, because the contract does not name materials....
- Practice · 8 marks A design engineer is asked to add fingerprint and voice recording to a security system, knowing buyers could use it to spy on s...

Model rules: obligations to society, employers and engineers

- QB B4(i) · 3 marks Describe a registered engineer's obligation to the following stakeholder group: (i) Society.
- QB B4(ii) · 3 marks Describe a registered engineer's obligation to the following stakeholder group: (ii) Employers and clients.
- QB B4(iii) · 3 marks Describe a registered engineer's obligation to the following stakeholder group: (iii) The engineering profession.
- CAT 1 (ethics-cat-1) Q1c · 2 marks each Describe the obligations of a registered engineer to each of the following stakeholder groups: (i) society, (ii) employers and ...
- 2020-21 exam Q1D · marks not stated Describe the obligations of a registered engineer to each of the following stakeholder groups: (i) Society, (ii) Employers and ...
- Slides 20, 22, 24 · practice State the conflict-of-interest clauses in the UIPE/ERB model rules.

Contract law and the problems of law in engineering

- QB C1 · 8 marks Madge's shop has been robbed. She says to Harold, a police officer: "If you find the thief, I will pay you $500." Harold, throu...
- 2017 Q4E · 5 marks List any three problems associated with application of the law in the field of Engineering.
- Practice · 4 marks State two benefits of law to the engineering profession and say when precise rules suit better than broad guidelines. (Original...

Codes of ethics: functions, limitations and canons

- 2016 Q2A · 4 marks What is a professional code of ethics?
- 2017 Q3A · 4 marks · see also t21 Describe what is meant by a Code of Ethics citing clear supporting examples.
- 2020-21 Q1A, QB B2 · 5 marks What is meant by a Code of Ethics (COE) and what are the known weaknesses of CoEs?
- 2017 Q3B · 6 marks Give four (4) reasons behind why these codes are important to society.
- 2015 Q1B · 2 marks · 2011/12 What is the purpose of a Code of Ethics in engineering practice?
- CAT 2021-22 Q1A (as worded in the solution file) What do you understand by a professional code of conduct and how is it important for any engineer?
- 2016 Q2D · 4 marks · 2017 Q4C · CAT 2021-22 Q1C List four fundamental principles of the Engineers' code of ethics (common to the different engineering society codes of ethics).
- CAT 1 2015 Q2C · 5 marks Highlight any 5 points of the IEEE code of ethics.
- QB B3 · 7 marks · 2020-21 Q1B · 2016 Q3B List the fundamental canons of the UIPE (Uganda Institute of Professional Engineers) Code of Ethics. Hint: similar to IEEE fund...
- 2013/14 paper · 8 marks [reconstructed from garbled scan] What are the common features contained in a code of ethics for engineers?

- 2016 Q2B · 10 marks · 2011/12 List any SIX roles/functions of professional codes of ethics.
- Practice · 4 marks Explain how a code of ethics supports an engineer, and how it deters misconduct.
- Practice · 5 marks Which function of codes of ethics is the most important, and which is the most open to abuse? Give reasons.
- 2016 Q2C · 6 marks · 2017 Q3C · 4 marks Describe any FOUR limitations of a professional code of ethics. (2017: Discuss two limitations associated with the use/applicat...
- CAT 2021-22 Q1D Discuss four limitations of the use of professional codes of conduct and use one of the ethical theories either to support or r...
- 2015 Q1D · 7 marks Imagine you are the president of a small high technology firm. Your company has grown over the last years to the point where yo...

IP, protection types, copyright and fair use

- QB D1 · 2 marks How is intellectual property different from tangible property?
- CAT 1 · Q2a · 2 marks What do you understand by the term intellectual property?
- CAT 1 · Q2b · 6 marks; also 2017 Q2D · 6 marks Differentiate between the different available methods of intellectual property protection.
- 2014/15 Q5c; also 2011/12 and 2013/14 papers · 3 marks Identify three (3) ways in which an engineer can protect intellectual property.
- 2016 Q4B · 6 marks What two forms of legal property protection are most frequently applied to software? List two or three of their key characteris...
- CAT 1 2021-22 solution file · Q3c · marks not stated Describe the key differences between intellectual property rights and copyrights.
- Solution file · CAT 1 2021-22 Q4a · marks not stated What justifies the need of a copyright policy in the Engineering profession?
- 2016 Q6 · 20 marks (A 10, B 5, C 5) Write a memo to the judge explaining A. the intellectual property claims Mofonic and Bunten can bring (10 marks), B. any defens...

Protection challenges, DRM and the landmark cases

- ethics-cat-1 Q2c · 6 marks With the aid of examples, briefly explain why intellectual property protection is much harder to enforce in this internet age.
- 2017 Q2C · 5 marks Briefly analyze why intellectual property protection is that much harder to enforce in the internet age. Illustrate your positi...
- 2014/15, 2013/14 papers · 6 marks How does Internet and other digital technology access undermine intellectual property protection? [reconstructed from garbled s...
- Practice · 4 marks Distinguish between time shifting and space shifting, naming the case in which each was held to be fair use.
- Practice · 4 marks What is reverse engineering of software, and when is it lawful? Name one case.

- Practice · 6 marks Explain the Napster and Grokster cases and what they decided about the liability of file sharing services.
- Practice · 5 marks Describe two examples of digital rights management used to protect CDs and DVDs, and the case that tested the DVD scheme.
- Practice · 6 marks Give three criticisms of digital rights management.

Copyright law, software protection and the DMCA

- Practice · 6 marks Outline the history of copyright law in the United States and give three criticisms of making the protection period longer.
- Practice · 6 marks Explain why the Digital Millennium Copyright Act is controversial. Give three consequences.
- Practice · 4 marks List the three US laws since 1997 that made punishments for copyright infringement harsher, and state what each provides.
- QB D2 · 6 marks What is the consequentialist justification for laws that give ownership and control of software to individuals or corporations ...
- 2016 Q4A · 8 marks; also CAT 1 2021-22 Q3b · utilitarian arguments on software property rights There are a number of arguments for and against strong property rights over computer software. Provide two utilitarian-style ar...
- 2016 Q4C · 6 marks; also exam 2021-22 solution file Q5d In a few sentences, explain why software poses a challenge to these traditional forms of intellectual ownership.
- Practice · 8 marks State what can and cannot be copyrighted in software, give three examples of copyright violation, and outline four criticisms o...

Open source and Creative Commons

- Practice · 6 marks What is open-source software? Explain the term copyleft.
- Practice · 9 marks Outline the benefits and the criticisms of open-source software.
- Practice · 4 marks What is Creative Commons? Give two examples of the licences it offers.
- Set 5 slide 36 · discussion A software service company A has a client C. C uses a software S made by company B. Company A made a copy of S, not to resell, ...
- Set 5 slide 36 · discussion Consider applying the fair-use guidelines to making a video of oneself lip-synching to a popular song and posting the video on ...
- Set 5 slide 37 · discussion Which is more likely to be effective in protecting intellectual property in digital media (e.g. CDs, DVDs): tougher copyright l...
- Set 5 slide 37 · discussion As the copying of digital texts, audio, and video grows cheaper, the restrictions needed to enforce copyright grow more expensi...

Safety, risk and acceptable risk

- 2017 Q5A · 3 marks Define the concept of safety as applied in the Engineering profession.

- CAT 2021-22 Q5a · marks not shown Describe your understanding of the safety concept as applied to the Engineer's discipline.
- CAT 2021-22 Exam Q6A · marks not shown Define safety and risk in respect to an engineering design and highlight differences between these two terms.
- 2017 Q5B · 4 marks Explain the term RISK and its relation to safety in Engineering.
- 2015 exam Q2a · 2 marks What do you understand by risk? [reconstructed from garbled scan]
- CAT 2021-22 Q5b · marks not shown Explain the term risk and how it is related to safety in Engineering.
- CAT 2021-22 Exam Q6C · marks not shown Under what circumstances would the risk in a given engineering design be considered acceptable and what factors affect this ris...

Costs of unsafe design, uncertainty and testing

- 2017 Q5C · 5 marks List any five costs associated with unsafe engineering designs.
- CAT 2021-22 Q5c · marks as set Describe any five costs that come with unsafe engineering equipment and/or design.
- 2017 Q5D · 3 + 2 marks · also CAT 2021-22 Q5e Explain why a clear understanding of the risks in a product design is necessary for (i) the product manufacturer (3 marks) and ...
- 2017 Q5E · 5 marks (also CAT 2021-22 Q5d) List any five questionable or clearly wrong practices by engineers over the course of their work that would negatively affect p...
- Set 6 slide question · marks as set "Safety in a commodity comes with a price." Explain.
- Practice · 8 marks Describe the uncertainties an engineer meets in the design process and the testing strategies used to deal with them.
- Exam 2021-22 Q6B · 4 strategies Describe four strategies that should be adopted to ensure that a given engineering design (product) meets the necessary safety ...
- Exam 2021-22 Q6D · 3 reasons Give three reasons that necessitate carrying out a risk-benefit analysis of an engineering design.
- QB Sec B Q2b · 14 marks If engineers are not afforded the luxury of seeking perfection, it follows that they must be willing to risk failure. Do you ag...
- 2015 Q2b, c · 2 + 6 marks · also QB Sec B Q2a, two papers · beyond the slides (b) Identify the four major elements of the risk management process. (c) Engineers seldom manage to eliminate risk in their pro...
- 2015 Q2d(i) · 2 marks · beyond the slides What do you understand by Strict Liability?

Privacy, the law and government surveillance

- Slides 2-3 · Practice · 3 marks What is the ethical conundrum in business and government collection of personal data?
- QB A2 · 8 marks What are the main ethical issues in sustaining a delicate balance between Privacy and Security issues when monitoring/censoring...

- 2013/14 paper Q3d · 5 marks Assuming that all the details of Marlinspike's account are correct, did Marlinspike act ethically in rejecting the request from...
- CAT 1 2021-22 Q3a Who are the stakeholders involved in the development, deployment, and use of smart lampposts?
- CAT 1 2021-22 Q3b Who should be consulted about such projects' goals and development?
- CAT 1 2021-22 Q3c Identify two dilemmas that you recognize from the above case study.
- CAT 1 2021-22 Q3d What steps might business and government teams need to take to access the information/perspectives needed to manage the ethical...
- CAT 1 2021-22 Q3e What are some ethical issues that should be addressed as part of the deployment of smart lampposts in cities? Do those issues d...
- CAT 1 2021-22 Q3f How might the deployment of smart lampposts be evaluated through the ethical lenses of rights, justice/fairness, utilitarianism...

Encryption and identity theft

- Practice · 4 marks Distinguish between public-key and private-key encryption.
- Practice · 6 marks Discuss the ethical issues raised by the use of strong encryption, considering both privacy and law enforcement.
- Practice · 4 marks How does spear-phishing differ from phishing? Name one other way identity theft is carried out.
- Practice · 5 marks Explain what is meant by identity theft. State the information that is stolen and describe two ways in which the problem can be...

Profiling, workplace monitoring, spam and surveillance technology

- QB D3 · 5 marks (also sol exam2122 Q4c) What ethical issues does the use of data mining raise with respect to privacy?
- QB A1(ii) · 5 marks Simon is a social media consultant. The CIO asks him to collect personal data on the staff in the IT department via Facebook an...
- 2014 and 2015 papers Q5(c) · 4 marks Identify two ways in which a company may violate an employee's right to privacy.
- QB D4 · 4 marks (also CAT 2021-22 Q4d) What ethical issues does the use of facial recognition systems raise with respect to privacy?
- QB D5 · 8 marks (also CAT 2021-22 Q4e, which asks for two reasons) "Honest, upright, law-abiding citizens have nothing to fear from the distribution of their personal data." Do you agree or not?...

Profession versus occupation, whistleblowing and conflict of interest

- ethics-cat-1 Q1a · 3 marks Explain the difference between an occupation and a profession.
- CAT 2021-22 Q1 (iii) / 2016 Q1c · part of 12 marks Differentiate between the following: ... Profession and Occupation.

- Exam 2021-22 Q1b · marks not printed With aid of examples briefly describe how profession differs from a mere occupation.
- 2017 Q5F · 3 marks What qualifies Engineering as a profession?
- QB B1 · 4 marks Give four characteristics that distinguish a professional body from other social organizations.
- Exam 2020-21 Q1C · marks not printed Which distinguishing features do professional bodies like the Uganda Institute of Professional Engineers (UIPE) have over other...
- ethics-cat-1 Q1b · marks not stated Discuss the factors that would motivate engineers to practice their profession while carrying out their day-to-day activities.
- 2013/14 Q4b and 2011/12 Q4b · 4 marks Distinguish between internal and external whistleblowing.
- 2013/14 Q4a and 2011/12 Q4a · 6 marks Identify and explain three (3) essential conditions for one intending to "blow the whistle" against his or her employer.
- 2013/14 Q4c and 2011/12 Q4c · 3 marks What do you understand by "critical loyalty" to an employer?
- 2013/14 Q4d · marks as printed, garbled Discuss the most likely impacts on the whistle blower in an organization after they have revealed information.
- 2011/12 Q4(d) · [reconstructed from garbled scan] (i) 3 marks; (ii) marks not shown (i) Identify Dr Hansen's ethical dilemma. (ii) Did Hansen do something unethical by insisting on speaking out? Support your ans...
- 2013/14 Q5a · 2 marks; 2011/12 Q5a · 2 marks What do you understand by "conflict of interest"?
- 2013/14 Q5b, 2011/12 Q5b, CAT 2 2013/14 Q1a, CAT 2 2014/15 Q1a · 2 to 4 marks Distinguish between actual and potential conflicts of interest.
- CAT 2 2014/15 Q1b and CAT 2 2013/14 Q1b · 2 to 4 marks Describe two unique situations in which an employee may experience an actual conflict of interest.
- 2013/14 Q5c and 2011/12 Q5c · 6 marks Identify three (3) unique situations that can present conflict of interest for an engineer.
- 2013/14 Q5(d) · 8 marks Engineer A is the part-time town engineer for Smithtown and also has a consulting engineering practice. [Case as above.] 1. Was...
- 2011/12 Q5(d) · 4 + 4 marks · [reconstructed from garbled scan] Maggie is in charge of the search for a firm to design and produce promotional posters and billboard ads when Harold Telecom la...

Ethics of the internet, AI, robots and the environment

- ethics-cat-1 Q3c · 6 marks Describe ethical issues that arise from the use of technologies like artificial intelligence for educational purposes. For each...
- 2017 Q2 A · 5 marks Internet data share and applications have become very much an integral part of our human day to day lives. List 5 benefits faci...

- 2017 Q2 B · 9 marks (same as CAT 2021-22 Q1.2 II) Describe three ethical issues that arise from internet access and usage. For each issue highlight the ethical dilemma therein i...
- QB C3 · 5 marks Robots raise several ethical issues especially concerning safety of human lives. Which ethical values should have been embedded...
- 2011/12, 2013/14, 2014/15 papers · 2 marks What do you understand by 'Sustainable development'?
- 2011-2015 papers · 4 marks Name two (2) forms of environmental pollution and explain one (1) problem associated with each.
- 2011-2015 papers · 6 marks Identify and explain three (3) principles that guide engineers in making environmental ethics decisions.
- 2011-2015 papers · 4 + 4 marks (i) Identify two (2) ethical problems associated with increasing the use of fossil fuel as a source of energy in the manufactur...
- 2013/14 paper Q8(a) · 3 marks What do you understand by the phrase "eco-redesign of a product"? [reconstructed from garbled scan; the scan reads "Fm-redesign...
- 2013/14 paper Q8(b) · 4 marks Mention any two ways in which a telecommunications/electrical engineer can contribute towards sustainable development in a coun...